# Chapter 2

# Data visualization in corpus-based contrastive linguistics and translation and interpreting studies

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While inferential statistics are indispensable in comparative corpus studies such as those conducted in contrastive linguistics (CL) and translation and interpreting studies (TIS), it is equally important for researchers to develop an intuitive understanding of the data. Visualizations serve as a key tool for this purpose. In this contribution we distinguish three types of comparative corpus data commonly encountered by contrastive linguists and translation scholars: language-specific data, undirected comparative data, and directed comparative data. We discuss three main visualization techniques: proportional visualizations, graph-based visualizations and ordination techniques. We explore the strengths and limitations of these approaches, point out some caveats and provide scripts (in R and Python) to enable readers to replicate the analyses.

## 1 Introduction

The availability of large amounts of multilingual corpus data has opened up new possibilities for empirical approaches to contrastive linguistics and translation and interpreting studies. For example, we can empirically determine the contexts in which comparable linguistic expressions from different languages are used (e.g. Atayan 2021b, Gast 2015, 2017a,b, 2019, 2022), and we can investigate the conditions under which specific expressions are translated in a specific way, (e.g. Marco 2018, Crible et al. 2019, Defrancq & Magnifico 2020, Defrancq 2021,



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Atayan 2021a, Yung et al. 2023). These possibilities come with new challenges, however. From the perspective of the researcher, it is sometimes hard to filter out more relevant from less relevant information, to understand the interplay of the variables used in a study, and to interpret the results in a theoretically meaningful way, beyond statements of the type "variable $x$ has a significant effect on variable $y$".

While inferential statistics are indispensable in the comparative analysis of corpora, it is important for researchers (and readers) to also develop an intuitive understanding of the data (cf. also Gries et al. 2020). Visualizations are a useful tool for this purpose. In this chapter we discuss the use of specific tools for visualizing bilingual data commonly encountered in contrastive linguistics and translation studies (cf. also Sönning & Schützler 2023 on the visualization of corpus data in general).

An obvious particularity of contrastive studies based on translation corpora is the relational nature of the data, which often consists of pairs of an original sentences and its translation or interpretation. While the original sentence is an instance of monolingual production, the translated sentence is determined both by the co(n)text[1] and the input from the source language. We thus distinguish between three ways of analyzing frequency distributions. First, before carrying out a comparative study we need to understand language-specific distributions, using common tools of corpus linguistics; second, we can compare distributions from different languages in a symmetric way ('undirected comparison'); and third, we can study the choices made in one language under the influence of another language, e.g. translations of a given source language expression in a target language ('directed comparison').

For illustration we use data analysed in previous studies by us (Atayan et al. 2019, Gast & Atayan 2019, Gast et al. 2019, Atayan 2021a,b) on what we call adverbials of 'immediate' or 'proximal posteriority'. These adverbials are introduced in Section 2, where we also provide a brief description of the data. In Section 3, we introduce techniques commonly used for categorical (count) data in monolingual studies, presenting some extensions for higher-dimensional data. These techniques can also be used for undirected comparative data, as is illustrated in Section 4. In Section 5, we demonstrate ways of visualizing translation data. Sankey diagrams are useful for this purpose, but have certain limitations. We therefore propose a new type of visualization, Context-Conditional Translation Graphs, which visualize translation choices made under specific contextual

[1] 'Cotext' is often used for the textual surrounding of a sentence and 'context' for the extralinguistic circumstances of production. We will use 'context' as a cover term for 'cotext' and 'context' narrowly defined in the following.

conditions. Section 5 moreover contains some comments on Bayesian networks, which can be used to factorize joint probability distributions, and to visualize these factorizations, enabling researchers to test models of bilingual language production. Section 6 contains the conclusions.

# 2 Background on the domain of inquiry and the data

## 2.1 Adverbials of immediate posteriority in French and German

In earlier work we studied ways of expressing immediate posteriority in several languages, including adverbials such as Engl. *immediately*, Germ. *sofort* and *gleich*, and Fr. *immédiatement* and *tout de suite* (Atayan et al. 2019, Gast & Atayan 2019, Gast et al. 2019, Atayan 2021a,b). We will use the abbreviation 'ImPost-adverbial' in the following. ImPost-adverbials locate an event *e* relative to some reference point *R* in such a way that *e* follows *R* without a relevant temporal gap. For example, in (1) Jane's arriving at home (*R*) and her going to bed (*e*) are temporally adjacent, which is here expressed by *immediately*.

(1) When [$_R$ Jane arrived at home] [$_e$ she *immediately* went to bed].

Adverbials like *soon* (cf. (2)), *bientôt* (French) and *bald* (German) are related to ImPost-adverbials insofar as their use implies a relatively short interval between events or points in time, but the interval is longer than in the case of ImPost-adverbials.

(2) *Soon* after [$_R$ she arrived], [$_e$ Jane went to bed].

We will refer to the class of adverbials covering both those of immediate posteriority and *soon*/*bald*/*bientôt* as 'adverbials of proximal posteriority', or ProxPost-adverbials for short. ImPost-adverbials are regarded as a special case of ProxPost-adverbials.[2]

Expressions of proximal posteriority tend to be context-sensitive, insofar as their use is (partially) determined by specific properties of the context. In a (simplifying) model of language production we can assume that there is a given function $f$ that a speaker wants to express – say, immediate posteriority – in some context $C$ with context variables $\{C_1 \ldots C_m\}$, each of which comprises a certain

[2]Note that arguably, there are also differences in the lengths of the intervals between, for instance, Germ. *sofort* (no or very short interval) and *gleich* (somewhat longer interval). We disregard these differences in the following, as they are not of primary interest to the questions addressed in our projects.

number of levels (e.g. 'directive', 'commissive' and 'representative' for the variable ILLOCUTIONARY FORCE). Moreover, there is a set of expression candidates $W = \{w_1 \ldots w_n\}$ which are compatible with function $f$. One aspect of use that we are interested in is the probablities with which the candidates $W = \{w_1 \ldots w_n\}$ are used in context $C$ to express function $f$. These probabilities can be regarded as conditional probabilities of the form $p(w|f, C)$: the probability of using expression $w$, given that we want to express the function $f$ in context $C$.

To illustrate some relevant context parameters determining the distribution of ImPost-adverbials, let us consider some examples. ImPost-adverbials may be used in utterances with representative illocutionary force, as in (1) above, or with directive illocutionary force, as in (3) below; they can be used anaphorically, with another event $e$ as a reference point (cf. (1)), or deictically, with the moment of speech as a reference point (cf. (3)); and they can be used with intentional (controlled) actions, as in (1) and (3), or with unintentional/spontaneous events, as in (4).

(3) [$_e$ Get out] *immediately*! (directive, deictic, intentional; implicit $R$)

(4) [$_e$ The house of cards collapsed] *immediately* when [$_R$ I touched it]. (representative, anaphoric, spontaneous)

ImPost-adverbials often exhibit particularities in their distribution amounting to interactions between their distributional properties. For example, Engl. *immediately*, while being compatible with both deictic reference points (as in (3)) and unintentional actions (as in (4)), is not normally used when these two context parameters co-occur, as in (5a), in a reading that is both spontaneous and deictic. This sentence is not normally understood as a warning about an imminent, spontaneous collapse, in which case we might utter something like (5b); rather, (5a) suggests that the collapse will follow some other event triggering it, as in (4) above and (5c) below.

(5) a. Be careful, it will collapse *immediately*!
    b. Be careful, it will collapse *any moment*! (deictic)
    c. Be careful, it will collapse *immediately* if you touch it! (anaphoric)

In translation and interpreting, there are, obviously, additional factors determining the probability of a specific form being used to express some function $f$ in a given context $C$. First, it is well-known that translated (or, more generally speaking, mediated) language production differs systematically from monolingual language production (cf. for instance the well-known translation universals investigated in the tradition of Baker 1993, 1995). Second, there are interference effects,

i.e., the choice of a target language form is partially determined by the form found in the input (often leading to "shining through" effects, cf. Teich 2003). There is, thus, an additional variable to be considered. The probability of using a given expressions is conditioned by both the context (type) $C$ and the source language expression $s$. We are thus dealing with a conditional probability of the form $p(w|f, C, s)$: the probability of using expression $w$, given that we want to express function $f$ in context $C$, with expression $s$ in the source text. For example, if we translate (5c) into German, we can determine probabilities for using *gleich* or *sofort* to express immediate posteriority in the specific context, given that the English original contains *immediately*. This probability may differ if the source sentence contained a different adverbial with a similar meaning, e.g. *right away*.

## 2.2 The data

For illustration we use two data sets, both of them from the Europarl corpus (Koehn 2005, Cartoni & Meyer 2012, Cartoni et al. 2013). The first data set was created in the context of a joint project on ImPost-adverbials carried out at the Universities of Heidelberg and Jena. The main results of this project have been published in Atayan et al. (2019), Gast et al. (2019), Gast & Atayan (2019), Atayan (2021a) and Atayan (2021b), to which the reader is refered for more information (e.g. concerning the annotation procedure). The data used for this chapter is based on two originally distinct data sets: (i) 508 annotated observations from German and French in which the ImPost-adverbials *gleich*, *sofort*, *immédiatement* and *tout de suite* occur either in originals or in translations, annotated by a team of annotators as mentioned in the aforementioned publications; and (ii) 100 observations each of the two adverbials of proximal posteriority *bientôt* (French) and *bald* (German), annotated by V. Atayan. The data as well as all R- and Python-scripts used for the analyses and visualizations are available in the OSF-repository https://osf.io/nufpv/.

The data were annotated for the following variables (the specific levels of the variables will be briefly explained when examples are discussed):

- Adverbial in original and translation
- French and German tense and mood
- Illocutionary force/modality
- Presence of a syntactic modifier
- Intentionality of the predicate
- Agent of the main predicate
- Reference point

# 3 Visualizing language-specific distributions

Even in a contrastive study, the first step consists in the inspection of the language-specific distributions. We mostly deal with four such distributions if we study two languages $L_1$ and $L_2$: $L_1$/original, $L_1$/translated (from $L_2$), $L_2$/original, and $L_2$/translated (from $L_1$). By inspecting these distributions we can gain a first impression of the differences between $L_1$ and $L_2$, as well as between the original and translated versions of both languages.

In this section we discuss three types of visualization techniques, summarized under 'proportional geometric visualizations' (Sect. 3.1), 'graph-based visualizations' (Sect. 3.2), and 'ordination techniques' (Sect. 3.3).

## 3.1 Proportional geometric visualizations

Among the techniques most commonly used for the visualization of monolingual data are what we can loosely call 'proportional plots', where frequencies observed in data are graphically represented in such a way that the size of some geometric shape is proportional to them. The most widely used plot types are probably mosaic plots and Cohen-Friendly association plots (Cohen 1980, Friendly 1982, 1994a). We will focus on mosaic plots, using the `vcd` package for R.[3] Consider Figure 1. The contingency table at the top shows how frequently the French ProxPost-adverbials *immédiatement*, *tout de suite* and *bientôt* are used with chronological or deictic reference points (i.e., with reference points in the surrounding text, as in (1), or with the moment of speech as the reference point, as in (3)). Each cell in the table corresponds to a combination of levels from the two variables ProxPost-adverbial and Reference Point. The rectangles in the mosaic plot at the bottom are proportional in size to the frequencies shown in the table at the top along both dimensions. Along the $x$-axis their length corresponds to the proportion of the relevant level within the category French ProxPost-adverbial, and along the $y$-axis the rectangles reflect the proportion of deictic and chronological observations. The mosaic plot shows, among other things, that the cells corresponding to chronological/*immédiatement* (top left) and deictic/*immédiatement* (bottom left), while exhibiting rather similar counts (32 and 29), exhibit very different relative frequencies in the rows for chronological and deictic cases.

Figure 1 also provides information about levels of deviation from statistical independence (a homogeneous distribution). The colour of a rectangle corresponding to a cell $C_{m,n}$ of a contingency table $C$ reflects the magnitude of the Pear-

[3] Mosaic plots can also be generated with the Python module `statsmodels`.

| | immédiatement | tout de suite | bientôt |
|---|---|---|---|
| chronological | 32 | 15 | 3 |
| deictic | 29 | 41 | 97 |

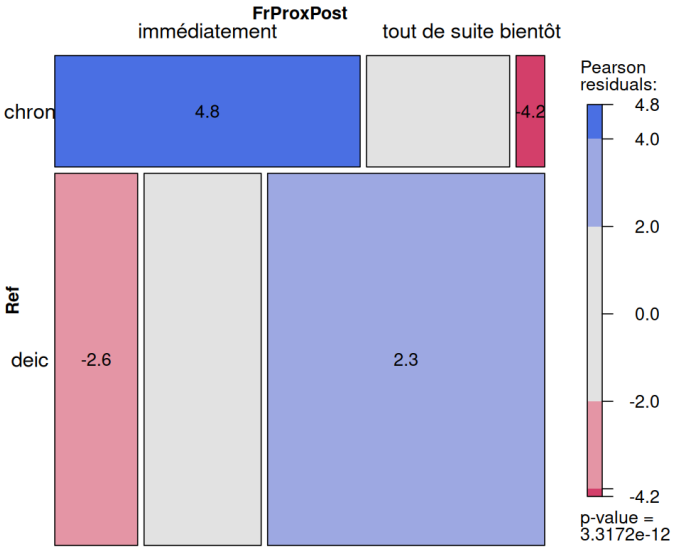



Figure 1: Contingency table (top) and mosaic plot (bottom) representing the frequencies as well as degrees of over- or underrepresentation for three French ProxPost-adverbial-adverbials relative to the variable REFERENCE POINT (*chron*: chronological, *deic*: deictic). The plot was created with the package `vcd` for `R` (Zeileis et al. 2007); cf. the script `01_proportional_visualizations.R` in the Supplementary Materials.

son residuals $E_{m,n}$ for $C_{m,n}$ (cf. Agresti 2002: Ch. 3 for details). Pearson residuals can be regarded as a standardized measure of deviation from statistical independence. They are calculated as the difference between the observed and expected frequency, divided by the square root of the expected frequency (see Sect. A.1 in the Appendix for some explanations). Pearson residuals approximately follow a standard normal distribution (but see below for a caveat), which means that we can estimate the probability of a cell count, given the assumption of statistical independence (homogeneity): $\approx$ 68% of the data are located within one standard deviation of the distribution, and $\approx$ 95% are located within two standard distributions. As we are typically interested in results beyond the 95% threshold ($\alpha = 0.05$,

for an undirected hypothesis), the thresholds 2 (for overrepresentation) and −2 for underrepresentation are particularly relevant.

As the legend to the right of the plot shows, the colours indicate the following magnitudes: light blue: $4 > E_{m,n} > 2$, dark blue: $E_{m,n} > 4$, light red: $-4 < E_{m,n} < -2$, dark red: $E_{m,n} < -4$. In cells with a significant over- or underrepresentation the Pearson residuals are moreover specified in the relevant rectangle. The plot suggests that *immédiatement* tends to co-occur with chronological reference points, while *bientôt* is attracted by deictic reference points. *Tout de suite* does not seem to show a particular association with either type of reference point.

An important point to note is that Pearson residuals do not exactly follow a standard normal distribution and "[provide] conservative indications of cells having lack of fit" (Agresti 2002: 81). Alternatively, an "adjusted" residual, which "results from dividing it by its standard error", can be used (Agresti 2002: 81). In the context of mosaic plots, Friendly (1994b: 7) notes that "the adjusted residuals are always larger in magnitude than the unadjusted values", and "adjusted residuals have the property that their standard errors are equal, so their values are more comparable over cells in the contingency table". He also holds that "experience shows that they provide *better visual display of the patterns of association* than do ordinary Pearson …residuals" (emphasis original).

As we do not focus on inferential statistics in this contribution, we will not go into any further details at this point (but see Sect. A.1 in the Appendix for some comments). Readers and users should be aware of the "meaning" of the colours in a mosaic plot, however, and take it into account in their interpretation of the data. In the following, we will continue to use unadjusted Pearson residuals for the sake of simplicity, because they also play an important role in ordination (where they are mostly called 'standardized residuals', see Sect. 3.3).

So far we have only crossed two variables. Figure 2 shows a mosaic plot where three variables have been crossed: (French) ProxPost-adverbial, Reference Point and Intentionality. In this case the (graphically) two-dimensional representation provided by a mosaic plot has its limitations, and the diagram effectively shows two plots, one at the top, where Intentionality is crossed with ProxPost-adverbial for chronological reference points, and one at the bottom, where this combination is shown for deictic reference points. We can see that three combinations have Pearson residuals higher than 2: {chronological, intentional, *immédiatement*}, {chronological, non-intentional, *immédiatement*}, and {deictic, non-intentional, *bientôt*}. {Deictic, non-intentional, *immédiatement*} and {deictic, non-intentional, *tout de suite*} and {chronological, intentional, *bientôt*} have residuals lower than −2.

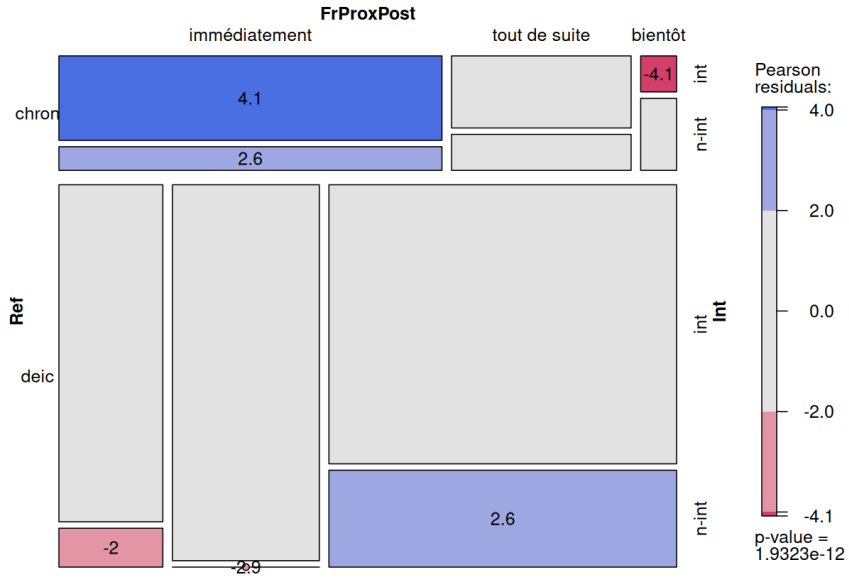



Figure 2: Mosaic plot representing the frequencies as well as degrees of over- or underrepresentation for three French ProxPost-adverbial-adverbials relative to the variables REFERENCE POINT (*chron*: chronological, *deic*: deictic) and INTENTIONALITY (*int*: intentional, *n-int*: non-intentional). The plot was created with the package `vcd` for `R` (Zeileis et al. 2007); cf. the script `01_proportional_visualizations.R` in the Supplementary Materials.

An alternative way of visualizing three-dimensional distributions is shown in Figure 3, in what we call a Feature-Combination Frequency Grid (FCFG) (see Gast 2017b). Each node in the grid corresponds to one feature combination, and the size of a sphere represents the observed or expected frequency of the relevant feature combination. The colours of the spheres reflect Pearson residuals, as in mosaic plots.

## 3.2 Context-Conditional Correlation Graphs

With Feature-Combination Frequency Grids (FCFGs) we can maximally visualize data along three dimensions. More dimensions are possible with 'stacked' mosaic plots as shown in Figure 2, but the information becomes increasingly difficult to process as the number of dimensions increases. In order to represent pairwise associations of variables in a single plot we can make use of graphs.

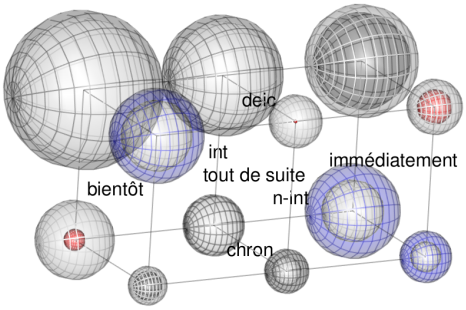



Figure 3: Feature-Combination Frequency Grid (FCFG) representing the frequencies as well as degrees of over- or underrepresentation for three French ProxPost-adverbials relative to the variables Illocutionary force and Intentionality (see Fig. 2 for the levels). The visualization was created using the package `rgl` for R; cf. the script `01_proportional_visualizations.R`.

We call graphs of the type shown in Figure 4 Context-Conditional Correlation Graphs (CCCG). The graph illustrates (significant) imbalances in the distribution of German ProxPost-adverbials relative to four variables (Intentionality, Agent, Reference point and Illocutionary force/Modality).[4] The arcs show, for each pair of variables, the levels for which there is a significant overrepresentation in a pairwise cross-tabulation, along with the Pearson residual (following the underscore). For example, the arc at the very top shows that in a cross-table $T$ of the variables ProxPost-adverbial and Agent, the Pearson residual for the cell $T_{3rd,bald} \approx 4$. The location of 'Ag3ps_4' (closer to *bald* than to *gleich*) indicates that this feature is overrepresented for *bald*.

The graph in Figure 4 only shows pairwise associations and, in a way, summarizes the information distributed over a large number of $2 \times 2$ cross-tables. In order to understand the interplay and mutual dependencies between the variables in a multivariate data set like ours, it often makes sense to use ordination techniques, to which we turn now.

[4]Variables and levels: Agent: *AgxPs* indicates that Agent is *x*. person; Illocutionary force/modality: *dir.boul*: directive/bouletic, *n.illoc*: not indicated, *poss*: possibility; Intentionality: *int*: intentional predicate, *n.int*: non-intentional predicate; Reference point: *chron*: chronological, *deic*: deictic; Modification: *modf*: adverbial is syntactically modified, *n-modf*: adverbial is not syntactically modified.

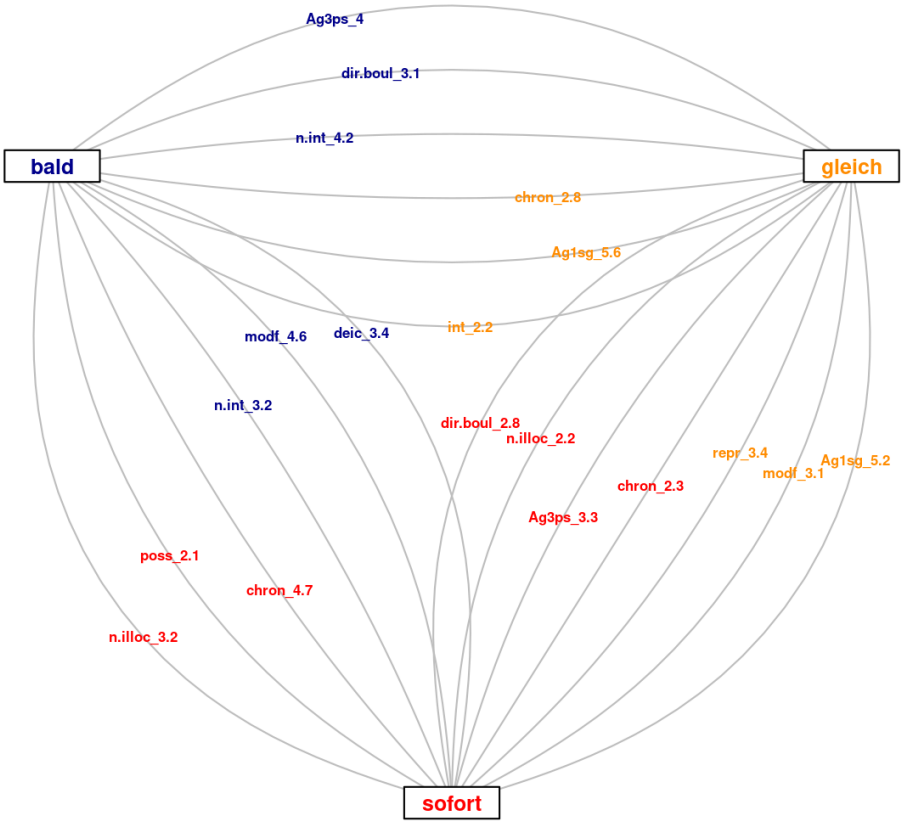



Figure 4: Context-Conditional Correlation Graph; the graph visualizes the distribution of German ProxPost-adverbials relative to four variables (Intentionality, Agent, Reference point and Illocutionary force; see Note 4 for the levels). It was created with the package `qgraph` for R (Epskamp et al. 2012); cf. the script `02_CCCG.R` in the Supplementary Materials.

## 3.3 Ordination

The term 'ordination' subsumes a number of methods that represent multivariate data in a low-dimensional space. In this section we discuss two types of ordination techniques that are useful for the analysis of categorical data, i.e., (simple symmetric) Correspondence Analysis (Sects. 3.3.1) and Multiple Correspondence Analysis (Sect. 3.3.2).

### 3.3.1 Simple symmetric Correspondence Analysis

Correspondence Analysis (CA) is probably the most basic type of ordination (see Greenacre 2016 for a comprehensive overview of the ordination techniques discussed in this section). It is based on the idea that contingency tables can be interpreted as coordinate systems. Consider the left-hand side of Table 1. The table shows rowwise relative frequencies of the German adverbials *gleich*, *sofort* and *unverzüglich* as translations of the French ImPost-adverbials *immédiatement* or *tout de suite* (so essentially it shows the distribution of German ImPost-adverbials in translations from French). Each row corresponds to a 'row profile', with the percentages adding up to 1. The columns represent three levels of the variable ILLOCUTIONARY FORCE/MODALITY, i.e., 'comm[issive]' (promises), 'dir[ective]/boul[etic]' (utterances with directive illocutionary force or bouletic modality, where the speaker expresses a desire, often amounting to an indirect directive), and 'repr[esentative]' (descriptions of the world).

Without providing any information about significant deviations from the independence model the row profiles illustrate associations between levels. For example, the table suggests that *gleich* is associated with representative illocutionary force, as 72% of its cases occur with that level, while for the other adverbials this column only shows 53% (*sofort*) and 22% (*unverzüglich*).

We can regard the columns of the left-hand side of Table 1 as constituting a coordinate system within which we can locate the three adverbials of interest. This is illustrated in Figure 5. The three axes originating at (0 0 0) (the origin) represent the three columns of the left-hand side of Table 1 and constitute a three-dimensional coordinate system. The adverbial *unverzüglich*, for instance, is located at the coordinate (0.11 0.67 0.22), corresponding to the third row profile in Table 1 (cf. the orthogonal lines emanating from the relevant point). The plot also shows the centroid of the three row points (see Sect. A.2 in the Appendix on how the centroid is calculated).

Table 1: Table with row profiles (left) and standardized residuals (right) for German ImPost-adverbials in translations from French, relative to the variable ILLOCUTIONARY FORCE

| | row profiles | | | standardized residuals | | |
|---|---|---|---|---|---|---|
| | comm | dir/boul | repr | comm | dir/boul | repr |
| *gleich* | 0.16 | 0.12 | 0.72 | 0.11 | -0.23 | 0.13 |
| *sofort* | 0.06 | 0.41 | 0.53 | -0.1 | 0.09 | -0.03 |
| *unverzüglich* | 0.11 | 0.67 | 0.22 | 0.01 | 0.21 | -0.17 |

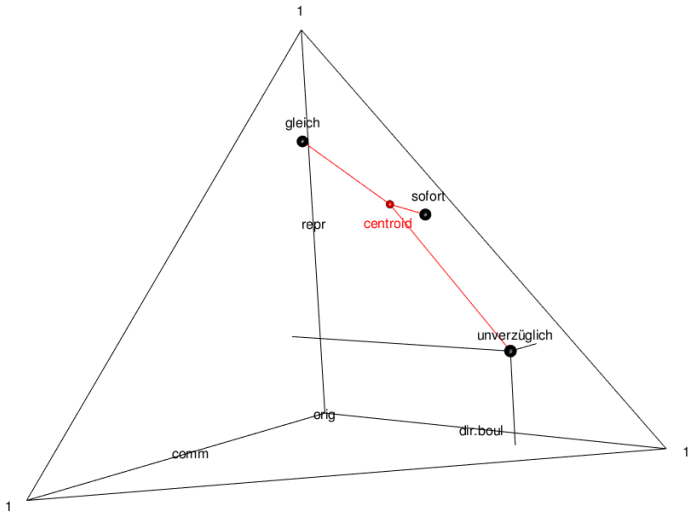



Figure 5: Visualizing Table 1, with the columns constituting a coordinate system for the rows. The plot was generated with the package `rgl` for R (Murdoch & Adler 2022).

The type of visualization shown in Figure 5 constitutes the basis for a lower-dimensional visualization using CA, which fits a new coordinate system to the data, in such a way that the largest amount of 'inertia' is captured on a low number of (so-called 'principal') axes – ideally two, as two axes can be easily visualized. Inertia is a measure of spread for count data around the independence model, corresponding to variance, which is a measure of spread around the mean value for continuous data. Inertia thus measures the degree to which a contingency table deviates from statistical independence. The term is inspired by the 'moment of inertia' in mechanics, which quantifies the distribution of mass around an axis of rotation.

CA proceeds as follows: first, the original contingency table (see Table 9 on p. 53) is converted into a percentage table, and then into a table of standardized residuals through division by the square root of the expected percentage (see the right-hand side of Table 1; cf. also Sect. A.2). As mentioned in Sect. 3.1, Pearson residuals indicate deviation from independence. The resulting matrix thus indicates to what extent a cell deviates from the independence model (assuming a homogeneous distribution). The centroid of the rows constituting this matrix is the origin of a three-dimensional coordinate system, which is rescaled along the three axes as a result of the standardization (division by the square root of the

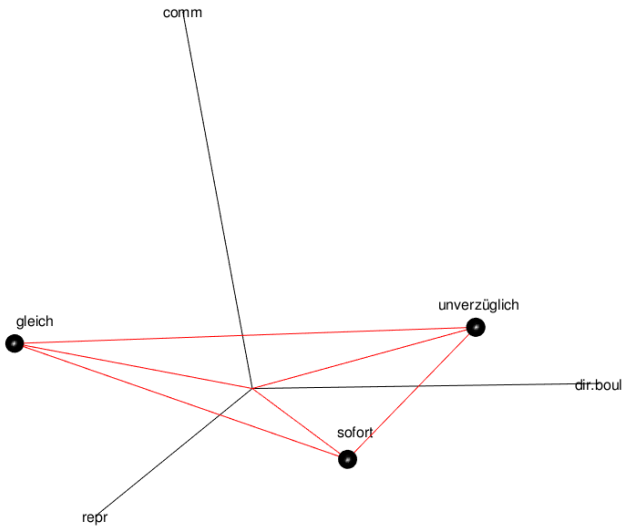



Figure 6: Visualizing the standardized residuals as shown on the right-hand side of Table 1. The plot was generated with the package `rgl` for `R` (Murdoch & Adler 2022).

expected frequency). The standardized residuals from Table 1 are visualized in Figure 6.

In a second step, the matrix of standardized residuals is subject to an operation called 'singular value decomposition' (SVD). The mathematical procedure of SVD is outlined in Sect. A.2 in the Appendix (based on Greenacre 2016). In the main text we will focus on the results.

SVD fits a new coordinate system to the data, in such a way that the first ('principal') axis captures most of the inertia, the second axis, which is orthogonal to the first, captures most of the remaining inertia, and so on until all data points can be precisely located in the newly created system. In the case of the data visualized in Figure 6, we can find a coordinate system with only two dimensions that preserves all the information contained in the table of standardized residuals, because three points are always located on a (two-dimensional) plane. If a table has been centered, as in the case of Pearson residuals, this plane moreover contains the origin (cf. the triangle in Figure 6).

The principal axes determined with SVD constitute the axes of a new coordinate system in which we can locate both the data points corresponding to the rows of the table of standardized residuals – in our case, the ImPost-adverbials *gleich*, *sofort* and *unverzüglich* – and the original axes corresponding to the levels

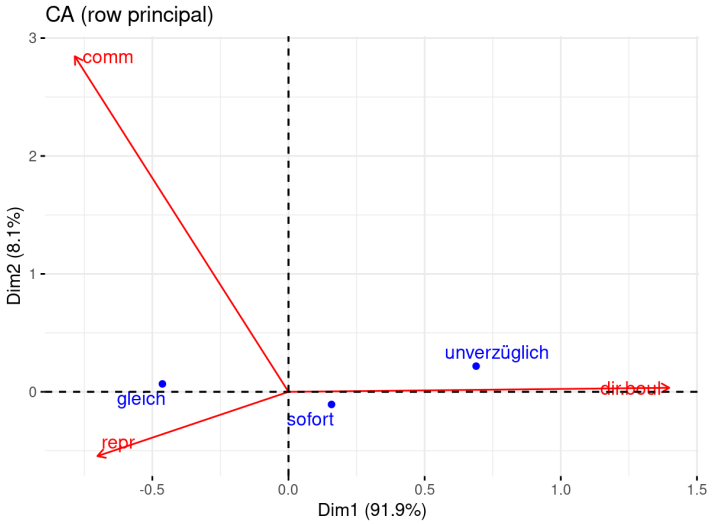



Figure 7: Correspondence Analysis for the data in Table 1. The CA was created with the package `ca` for `R` (Nenadic & Greenacre 2007). For the visualization we used the package `factoextra` (Kassambara & Mundt 2020). The plot shows a 'row analysis', with row points in principal coordinates and column vectors in standard coordinates.

of the variable Illocutionary Force ('comm', 'dir.boul', 'repr'). A CA created with the package `ca` for `R` (Nenadic & Greenacre 2007) is visualized in Figure 7.[5]

Along with the two-dimensional representation of the standardized residuals Figure 7 shows percentages on both axes. These percentages indicate how much inertia the relevant axis covers. Inertia has a direct connection to the Chi-squared test: it is identical to the Chi-squared statistic, divided by the number of observations; it is also the square root of the $\phi$-coefficient, a correlation measure for contingency tables; and it relates directly to the Chi-squared distances of the row coordinates to the origin. The Chi-squared distance is defined as the square root of the sum of the squared differences between the observed values and the expected values, divided by the expected values. It is a measure of deviation from independence. The inertia of the table is the sum of the Chi-squared distances squared, and weighted by the row masses.

Consider Figure 8 for illustration. The left-hand side shows the squared distances of the three coordinates representing the ImPost-adverbials to the origin, corresponding to the inertia (without scaling by the row masses). The right plot

[5]In Python, the module `prince` can be used for CA.

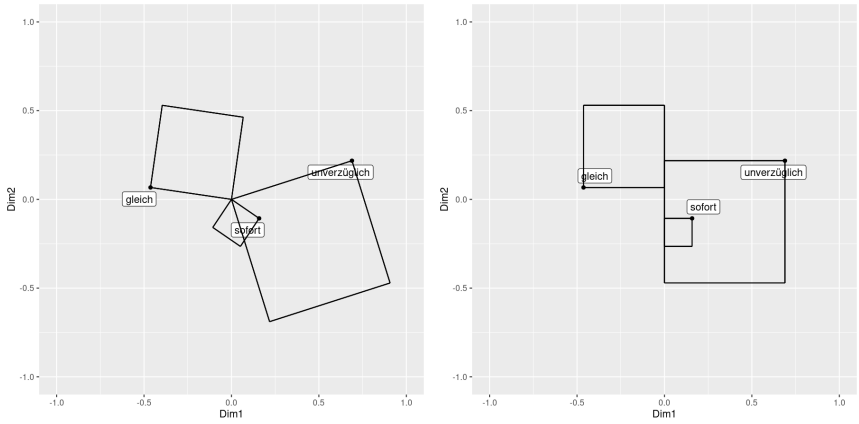



Figure 8: Squared Chi-squared distances (left) and squared Dimension 1 coordinates (right). The plot was generated with the packages `ggplot2` and `ggrepel` for `R` (Wickham 2016, Slowikowski 2024).

shows squares based on the lines that run parallel to the x-axis (the first principal axis) from the three ImPost-adverbials (also unweighted by the row masses). The area covered by the squares in the right plot amounts to 91.9% of the area covered by the total inertia as shown in the left plot. This is what the percentage on the x-axis in Figure 7 indicates.

The inertia not only gives us information about the degree of truthfulness of the dimensionality-reduced representation relative to the original, higher-dimensional one, it also tells us how 'important' specific axes are. When looking at Figure 7 one might be led to believe that the two axes explain a similar amount of inertia; but Axis 1 alone explains ≈ 92% of it. Note that all adverbials as well as the two levels of ILLOCUTIONARY FORCE ('repr', 'dir/boul') are very close to the $x$-axis – with low values on the $y$-axis –, implying that the $y$-axis explains a low amount of their row inertia.

So far we have focused on the position of the points representing the rows (ImPost-adverbials) in what is called a "row analysis" of CA. Remember that our starting point was the location of ImPost-adverbials relative to distributional variables, which constitute a coordinate system within which we can locate the latter. In principle, CA is symmetrical and we could just as well locate columns relative to rows. In fact, there are many ways of visualizing CAs, implemented, for instance, in the `fviz_ca_biplot` function of the `factoextra` package (Kassambara & Mundt 2020), or the `plot.ca` function of the `ca` package for `R` (Nenadic & Greenacre 2007). In a row analysis the rows (ImPost-adverbials) are

represented in so-called 'principal coordinates', which preserve Chi-squared distances between the observations. The columns, represented as vectors in Figure 7, are displayed in 'standard coordinates', which are so-called because they have a mean value of 0 and a variance of 1 (Greenacre 2016: 62).

In this display the relationship between an ImPost-adverbial (row) and a variable (column) is not represented by the distances between row and column vectors; it (ideally) corresponds to the inner products of these vectors ("ideally" bacause information may be lost as a result of the dimensionality reduction). This means that the degree of association between a row and a column is a function of the degree to which the two vectors point in the same direction. If all of the information has been preserved, we can reconstruct the values of the original table from the map shown in Figure 7. For example, the standardized residual for the combination *gleich*/'repr' is the inner product of the two vectors for *gleich* (scaled by the square root of its row mass) and the vector pointing at 'repr' (scaled by the square root of the average profile; see Greenacre 2016 for more information and Sect. A.2 in the Appendix for the calculation of this example).

While not everybody may be interested in the mathematical details, it is important to be aware of the specific type of map that is used for visualization. The 'row principal' representation chosen in Figure 7 is rather transparent, but often has the disadvantage of a visual imbalance between the row points, which are closer to the centre, and the column points, which are spread out more. Users should be aware, however, that a representation that condenses the data – while visually being more pleasing – potentially does not have the inner product interpretation of an asymmetric row principal plot, where the rows are represented in principal coordinates, and the columns in standard coordinates (see Greenacre 2016: Ch. 13 for discussion).

Representations like the one in Figure 7 give us an idea of the relationship between the levels of two variables. However, with a multivariate data set like ours, we need to be able to represent relationships between more than one variable. A very simple way of visualizing more than one variable is provided by a 'stacked' CA (Greenacre 2016: Ch. 17). We can simply concatenate contingency tables (horizontally or vertically) and carry out a CA on that table. Horizontal concatenation is illustrated for our data in Table 2, which shows the percentages for the variable Illocutionary force on the left, and for Agent on the right.

Given that the number of observations per variable is invariant, the Chi-squared distances between the levels of each variable are preserved in this type of stacked CA. The positions of the various levels relative to each other are thus not affected by the stacking (cf. Figure 9). However, it is clear that some information is lost in a stacked CA, as we do not get any information about the

Table 2: Stacked contingency table for German ImPost-adverbials (in translations from French) and the explanatory variables Illocutionary force and Agent

| | comm | dir.boul | repr | 1pl | 1sg | 2ps | 3ps |
|---|---|---|---|---|---|---|---|
| *gleich* | 0.08 | 0.06 | 0.36 | 0.06 | 0.22 | 0.06 | 0.16 |
| *sofort* | 0.03 | 0.21 | 0.26 | 0.07 | 0.01 | 0.06 | 0.35 |
| *unverzüglich* | 0.06 | 0.33 | 0.11 | 0.06 | 0.00 | 0.00 | 0.44 |

relationship between the levels of the explanatory variables (here, Illocutionary force and Agent). Table 2 suggests that a certain number of ocurrences of *gleich* have commissive illocutionary force, and another set of observations has a first person agent. The table does not tell us how many of the commissive observations have first person agents. In other words, the distributional variables are treated as being independent. While this is a clear drawback of the stacked CA, the method can be useful in order to determine the degree of independence of predictor variables. In particular, it can be instructive to compare a stacked CA to a Multiple Correspondence Analysis, to which we turn next.

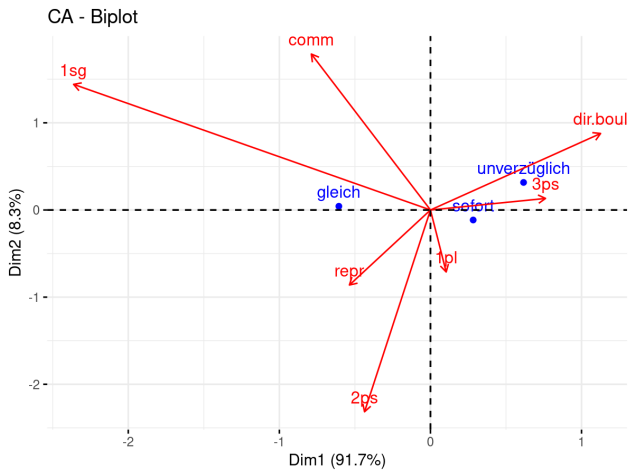



Figure 9: Stacked Correspondence Analysis showing the location of three German ProxPost-adverbials in a coordinate system derived from their associations with the variables Illocutionary force and Agent. The CA was fitted with the package `ca` for `R` (Nenadic & Greenacre 2007) and the plot was created with the package `factoextra` for `R` (Kassambara & Mundt 2020).

### 3.3.2 Multiple Correspondence Analysis

From a mathematical point of view, Multiple Correspondence Analysis (MCA) functions very much like (simple) CA. However, it is based on a representation of the data that takes the relationships between all (explanatory as well as response) variables into account. There are two (near equivalent) ways of accomplishing this: (i) carrying out a CA on the indicator matrix of a data set, or (ii) using the Burt table instead.

An indicator matrix is a binary matrix that indicates whether or not an observation exhibits a given property. Consider Table 3, which shows a random fragment of our data frame (German translations of French ImPost-adverbials). Each row corresponds to one example, and the columns specify the illocutionary force, the Agent, and the German adverbial.

Table 3: Random fragment of the data frame

| ID | Illoc | Agent | German ImPost-adverbial |
|---|---|---|---|
| 214 | repr | 1sg | *sofort* |
| 216 | dir.boul | 1pl | *sofort* |
| 217 | repr | 3ps | *gleich* |
| 218 | repr | 3ps | *sofort* |
| 221 | dir.boul | 3ps | *unverzüglich* |

The indicator matrix corresponding to Table 3 is shown in Table 4. Here, each column corresponds to a level of a variable, not a variable. While Table 3 specifies the levels themselves in the cells, Table 4 shows whether (1) or not (0) the observation in question is an instance of the relevant level.

Table 4: Indicator matrix corresponding to Table 3

| | Illoc | | | Agent | | | | ImPost-adverbial | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | comm | dir.boul | repr | 1pl | 1sg | 2ps | 3ps | *gleich* | *sofort* | *unvzgl.* |
| 214 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 215 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 |
| 216 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |
| 217 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| 218 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |

A Burt matrix is a cross-tabulation of the frequencies of all levels of all variables with all other levels of all other variables. Table 5 shows a fragment of our data in the form of a Burt matrix. Burt matrices and indicator matrices are sys-

Table 5: Burt matrix corresponding to Table 3

| | *gleich* | *sofort* | *unverzüglich* | comm | dir.boul | repr |
|---|---|---|---|---|---|---|
| *gleich* | 25 | 0 | 0 | 4 | 3 | 18 |
| *sofort* | 0 | 34 | 0 | 2 | 14 | 18 |
| *unverzüglich* | 0 | 0 | 9 | 1 | 6 | 2 |
| comm | 4 | 2 | 1 | 7 | 0 | 0 |
| dir.boul | 3 | 14 | 6 | 0 | 23 | 0 |
| repr | 18 | 18 | 2 | 0 | 0 | 38 |

tematically related insofar as the Burt matrix is the inner product of the transpose of the indicator matrix with itself ($B = I^{\mathsf{T}}I$, for an indicator matrix $I$).

While MCA is "mathematically" quite similar to CA, from the point of view of interpretation the two techniques are quite different. In CA, the rows correspond to categories of interest and the columns to the levels of a variable. In MCA, the rows correspond to individual observations (in the indicator matrix) and the columns correspond to variables that the individuals do (not) exhibit. Unlike CA, MCA thus provides information about individual observations. Moreover, in MCA *all* (levels of) variables are related to each other via Chi-squared distances.

A downside of MCA is that the underlying data tables – the indicator matrix (Table 4) or the Burt matrix (Table 5) – are highly redundant. The indicator matrix contains a lot of zeros, which turn into negative values when the residuals are calculated. The Burt matrix contains a lot of uninformative material on the diagonal, where levels are crossed with themselves. Because of the redundancy, both tables inflate the overall inertia. As a result, MCAs typically appear to explain a low percentage of inertia on their first principal axes. Greenacre (2016) has proposed a way of 'adjusting' the inertia in MCAs, which is implemented in the package `ca` for `R` (Nenadic & Greenacre 2007).

Figure 10 shows an MCA for German translations of French ImPost-adverbials, with three variables: ImPost-adverbial, Illocutionary force and Agent.[6] Without adjustment, the percentage of inertia captured by the first two principal axes is 26.3% (Axis 1) and 19.2% (Axis 2). It is much higher when computed with Greenacre's (2016) adjustment (72.7% [Axis 1] and 11.0% [Axis 2]).

An MCA creates a new coordinate system that is structured by all variables relative to all other variables. It creates a mapping from frequency distributions to positions in this system. We can therefore also locate variables (and observations)

[6]We used the package `FactoMineR` for `R` (Lê et al. 2008). In Python, the module `prince` can be used for MCAs.




Figure 10: Multiple Correspondence Analysis for the variables Illocutionary force, Agent and ImPost-adverbial (German translations of French ImPost-adverbials). The MCA was fitted with the package `FactoMineR` for `R` (Lê et al. 2008), the plot was generated using the package `factoextra` (Kassambara & Mundt 2020).

in MCA-coordinates without them being used to create those coordinates. In MCA speech, this means that we use 'supplementary' (or 'passive') variables. In a data set like ours it often makes sense to treat the target variable – in our case, expressions of immediate or proximal posteriority – as a supplementary variable. The idea is that the MCA-coordinates are defined by explanatory variables only, and that the adverbials in question can be located in terms of these coordinates. This is shown in Figure 11 for the same data that was used to generate the stacked CA in Figure 9, and the MCA without supplementary variables in Figure 10.

## 3.4 Comparing CA and MCA

The three ordination methods mentioned above provide ways of displaying different types of interdependencies between variables. A stacked CA represents the data in a coordinate system derived from the association between rows and columns, and positions the rows (adverbials) relative to the columns (distributional variables), or vice versa (the method is symmetric, although our focus has been on the row analysis). CA visualizes relations between context variables and adverbials, but not relations among the context variables themselves.

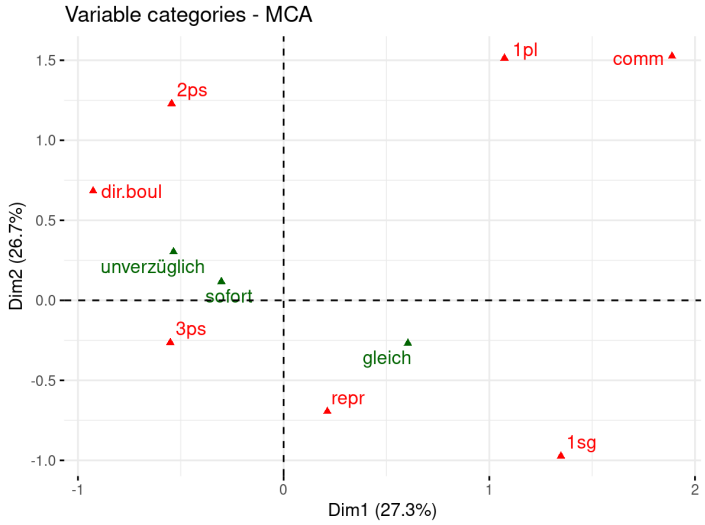



Figure 11: MCA for the variables Illocutionary force and Agent, with ImPost-adverbial as a supplementary variable (German translations of French ImPost-adverbials).) The MCA was fitted with the package `FactoMineR` for `R` (Lê et al. 2008), the plot was generated using the package `factoextra` (Kassambara & Mundt 2020).

An MCA with supplementary variables, in a sense, emphasizes the opposite perspective: it constructs a space based on the relationships among the context variables, and then projects the adverbials into this space. An MCA without supplementary variables represents relationships among all variables simultaneously, i.e., it does not distinguish between context variables and adverbials.

The differences between the three types of visualizations can be illustrated with the relationship between one German adverbial, *unverzüglich*, and two context variables, Illocutionary force and Agent. Consider first the (stacked) CA shown in Figure 9. The ImPost-adverbial *unverzüglich* and the category '2nd person' are located on opposite ends of the origin, suggesting a negative association. As a matter of fact, the variables are complementary, in the sense that *unverzüglich* is never used with the second person in our sample. In Figures 10 (MCA without supplementary variables) and 11 (MCA with supplementary variables), the second person is located in the same quadrant as *unverzüglich*. The reason is that both categories are distributionally related to directive illocutionary force or bouletic modality ('dir.boul'), though not to each other. In a way, the MCAs suggests a deceiving distributional association between *unverzüglich* and the second person. What this example illustrates is that none of the three

methods considered above – stacked CA, MCA with or without supplementary variables – can, in itself, provide a complete picture of the data. It is therefore instructive to experiment with different methods in order to develop a better understanding of the data. This does not mean, of course, that readers should feel encouraged to cherry pick those plots that they like best. Remember that the primary focus of this contribution is to support researchers in understanding their data, not in applying inferential statistics.

## 4 Visualizing undirected comparative distributions

In order to compare two languages in terms of the way a given functional domain is structured, a common procedure is to use a semantic or functional criterion as a third of comparison, and then determine the way the semantic space in question is mapped to linguistic expressions (Gast 2012). In our case, this means that we use the functional annotations as a *tertium comparationis*, and we locate the expressions from different languages relative to these annotations.

In Sect. 3 we used four methods to visualize form-function mappings in specific languages: CA – with and without stacked tables – and MCA – with and without supplementary variables. In this section we focus on only one of these methods, MCA with supplementary variables. The advantage of an MCA over a CA (with stacked tables) is that it allows us a per-item analysis, as each observation, and not just the variables, are located in low-dimensional space. The advantage of an MCA with supplementary variables is that it creates a coordinate system that is independent of the specific expressions used in a language, so we can (comparatively) locate expressions from different languages in that system. Put differently, the coordinate system is constituted by the distributional variables, functioning as the *tertium comparationis*, while the expressions are the *comparanda*.

An MCA with 'IP-adverbial' as a supplementary variable is shown for original occurrences of *gleich*, *sofort*, *immédiatement* and *tout de suite* in Figure 12, relative to four distributional variables (INTENTIONALITY, AGENT, ILLOCUTIONARY FORCE and REFERENCE POINT).

Though no doubt quite informative, Figure 12 is hard to read. A general observation that stands out is that the four adverbials are located in different quadrants of the plot. Another relevant observation is that *tout de suite* is considerably closer to the centre than the other adverbials, which suggests a less specialized distribution for this adverbial.

One way of simplifying a multivariate distribution as shown in Figure 12 is to apply clustering. As an MCA locates every single observation in the new space

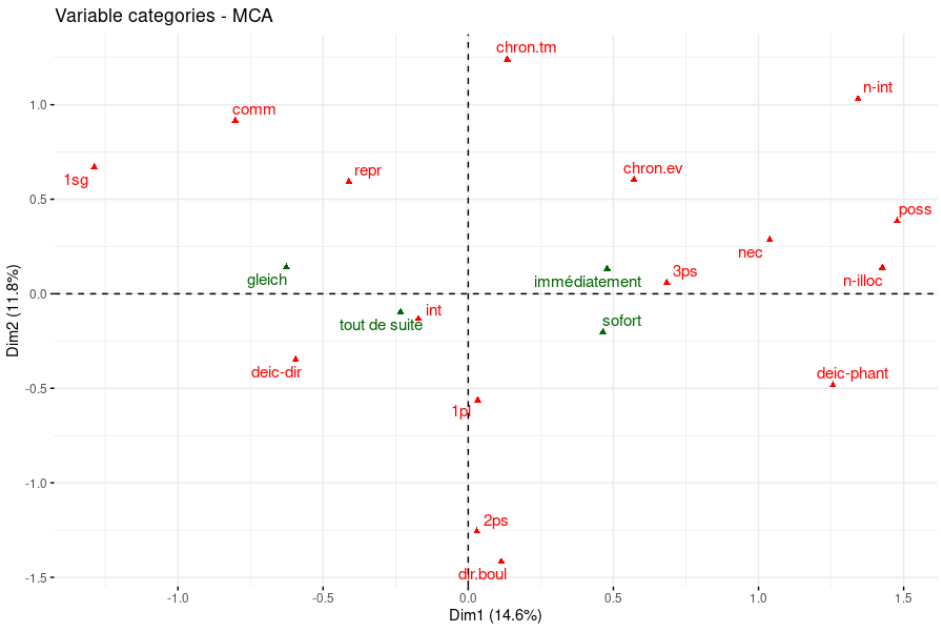



Figure 12: MCA with IP-adverbial as supplementary variable (German and originals), relative to the variables 'intentionality', 'agent', 'illocutionary force' and 'reference point'. The CA was created with the package `FactoMineR` for `R` (Lê et al. 2008), for the visualization we used the package `factoextra` (Kassambara & Mundt 2020). The adjusted percentages of explained inertia as returned by the package `ca` for `R` (Nenadic & Greenacre 2007) are 48.5% (Axis 1) and 15.0% (Axis 2).

that it creates, we can identify clusters of similar observations. We used PAM (partitioning around medoids) clustering for this purpose, which identifies clusters on the basis of distances between observations – as the name suggests, around a medoid, i.e., a (representative) observation in the centre of the cluster. [7] Clusters with 3 or 6 medoids were chosen because the average silhouette scores showed visible leaps at these points (in comparison to solutions with 2 and 5 clusters; see Figure 25, Sect. A.3, in the Appendix). The location of clusters of individual sentences in the coordinate system defined by the MCA is visualized in Figure 13. The annotations for the medoid observations of the clusters are shown in Table 6. The corresponding examples are provided in (6)–(8).

[7] We used the package `cluster` for `R`, cf. Maechler et al. (2023). We validated the clustering solution using the silhouette method (see the package documentation for more information).

Table 6: Medoid observations for cluster solutions with 3 (left) and 6 (right) clusters

| ID | agent | int | mod | ref | cl | ID | agent | int | mod | ref | cl |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 448 | 1sg | int | repr | deic-dir | 1 | 448 | 1sg | int | repr | deic-dir | 1 |
| 16 | 3ps | int | repr | chron.ev | 2 | 41 | 3ps | int | repr | deic-dir | 2 |
| 443 | 3ps | int | dir.boul | deic-dir | 3 | 443 | 3ps | int | dir.boul | deic-dir | 3 |
| | | | | | | 441 | 3ps | int | repr | chron.ev | 4 |
| | | | | | | 455 | 3ps | int | dir.boul | chron.ev | 5 |
| | | | | | | 280 | 3ps | n-int | repr | deic-phant | 6 |

(6) a. Je lui donne bien volontiers tout de suite la parole. (original)

b. Ich erteile ihm daher gern sogleich das Wort.

c. I am very happy to give you the floor now.

(7) a. C'est une expérience amère, lorsque l'on fait des propositions de réforme, de constater que d'autres les bloquent constamment avec des référence à autre chose.

b. Ich finde es bitter, wenn man Reformvorschläge macht, dass die gleich immer so abgeblockt und mit etwas anderem in Bezug gebracht werden. (original)

c. It is a bitter experience, when one makes reform proposals, to find that others are constantly blocking them with reference to something else.

(8) a. Il faut mettre ce régime sous tutelle internationale et il faut le faire tout de suite!

b. Dieses Regime muß unter internationale Aufsicht gestellt werden, und zwar sofort! (original)

c. This regime needs to be placed under international supervision, and this should be done straight away!

In the French and German versions of the medoid of Cluster 1 (cf. (6)) we find *tout de suite* and *sogleich*, arguably a stylistic variant of *gleich*. The medoid of Cluster 2 (cf. (7)) also has *gleich* in German but no expression in French – a not uncommon situation. The medoid of Cluster 3 (cf. (8)) has *sofort* in German and *tout de suite* in French. This illustrates that the clusters are not actually constituted by German or French expressions. They represent characteristic utterance types defined by the distributional variables: (i) a (typically deictic) declaration of future action by the speaker (Cluster 1); (ii) a (typically chronological) statement about a third person (Cluster 2); and (iii) a call for action (Cluster 3).

In order to understand associations between clusters (major use types) and ImPost-adverbials we can cross-tabulate the ImPost-adverbials with the clusters

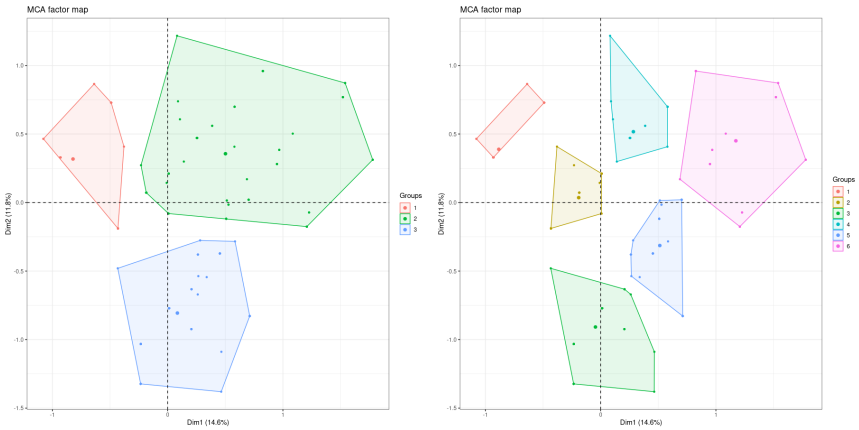



Figure 13: Clusters based on MCA-coordinates and PAM-clustering with three (left) and six (right) clusters, using the package `cluster` for `R` (Maechler et al. 2023). The plots were generated with the package `factoextra` for R (Kassambara & Mundt 2020).

of the two cluster solutions. Relevant mosaic plots are shown in Figure 14. Both plots suggest a positive association of Cluster 1 with *gleich*. The plot with six clusters moreover shows positive associations between *immédiatement* and Cluster 6, and between *sofort* and Cluster 5. Another important observation is that *tout de suite* does not show an association with any of the clusters. This confirms our impression that it is the most generic/least specialized expression of the ImPost-adverbials under investigation.

The visualization methods that can be used for undirected comparative data are thus essentially the same as those that were discussed for language-specific

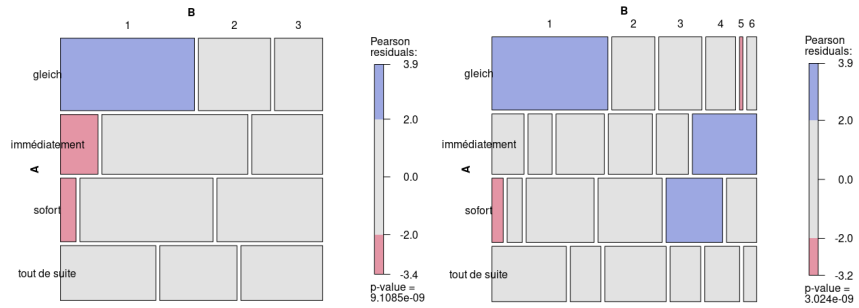



Figure 14: Mosaic plots for cross-tabulations of German and French ImPost-adverbials with clusters, for cluster solutions with three (left) and six (right) clusters. The plots were generated with the package `vcd` for `R` (Zeileis et al. 2007).

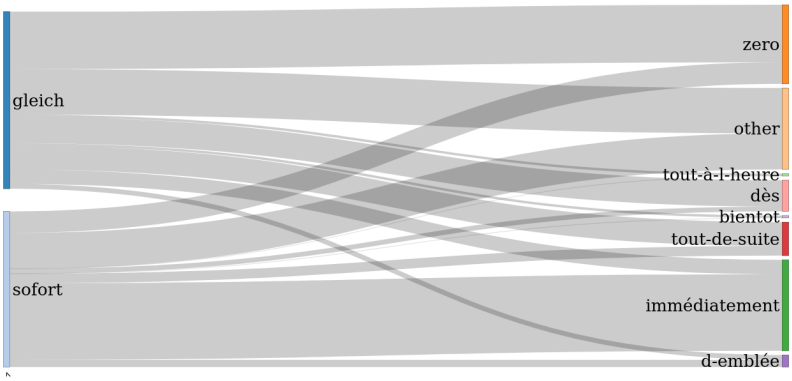



Figure 15: Sankey plot showing French translations of the German ImPost-adverbials *gleich* and *sofort*. The plot was generated with the package `networkD3` for `R` (Allaire et al. 2017).

data. What distinguishes comparative data from language-specific data is primarily the use of an appropriate *tertium comparationis*. This is a conceptual rather than technical matter. Note also that – as was pointed out above – in undirected comparative (e.g., contrastive) studies, it is always helpful to visualize language-specific distributions as well. For example, it is of course conceivable for context variables to behave differently (e.g., in their clustering behaviour) across languages. Such differences can best be identified by inspecting and comparing language-specific distributions, potentially identifying hypotheses that can be tested with inferential statistics.

# 5 Visualizing directed comparative distributions

## 5.1 Sankey diagrams and translation graphs

One of the simplest ways of representing translation choices visually is by using Sankey diagrams, named after H.R. Sankey (Kennedy & Sankey 1898). Figure 15 (generated with the package `networkD3` for `R`, cf. Allaire et al. 2017) shows a Sankey plot visualizing the frequencies of (French) target language choices for the two German source language expressions *gleich* and *sofort*.[8]

We can add more information to 'plain' Sankey diagrams by including an intermediate layer of nodes displaying the frequencies of the levels of some variable relative the source and target expressions, as in Figure 16. The plot shows

[8] In Pyhon, the `plotly` module can be used to create Sankey diagrams.

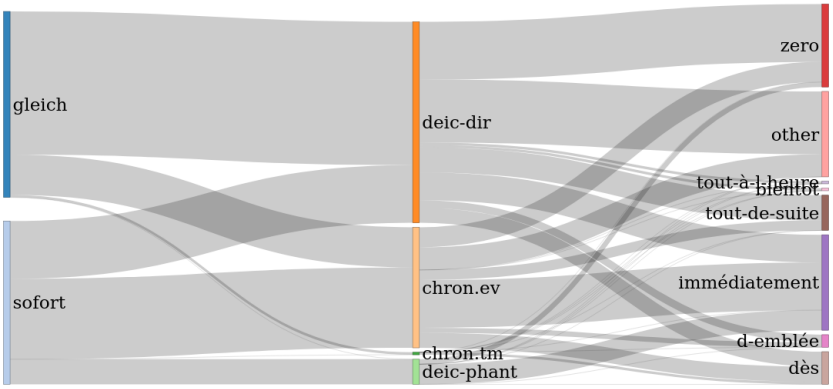



Figure 16: Comparative Sankey plot showing associations of IP-adverbials with French IP-adverbials, given the value for the variable Reference point. The plot was generated with the package `networkD3` for `R` (Allaire et al. 2017).

the frequencies of source and target language correspondences mediated by the variable Reference point. We can see that most cases of 'deic[tic]-dir[ect]' come from *gleich* and go to 'zero' or 'other' (in French), while almost all cases of 'deic[tic]-phant[asma]' come from *sofort* and go to *immédiatement*. Cases of 'chron[ological]-ev[ent]' seem to be typically associated with *sofort* in German originals, and with *immédiatement* in French translations.

To obtain a more global view, consider Figure 17, which shows the correspondences between source and target language expressions relative to the clusters mentioned in Sect. 4 (solution with three clusters). We already know that Cluster 1 is positively associated with *gleich* in German. The plot shows that most cases of this category go to 'zero' or 'other' in French. Clusters 2 and 3 are more evenly distributed among *gleich* and *sofort* in German, but they also seem to exhibit clear asymmetries.

While Sankey plots contain useful (relational) information about frequency distributions in the source and target language, they do not contain by-item information and are potentially deceptive. The edges conecting the nodes reflect frequencies, but the left edges are not connected by-item to the right edges. For example, as mentioned above, if we look at the middle node 'chron.ev' (for 'chronological relative to an event'), the plot suggests that those cases that are rendered with *gleich* in German go to 'zero' and 'other' in French. While this is probably true, the underlying data does not actually contain that information. Like a CA, Sankey diagrams are built on cross-tables and merely represent fre-

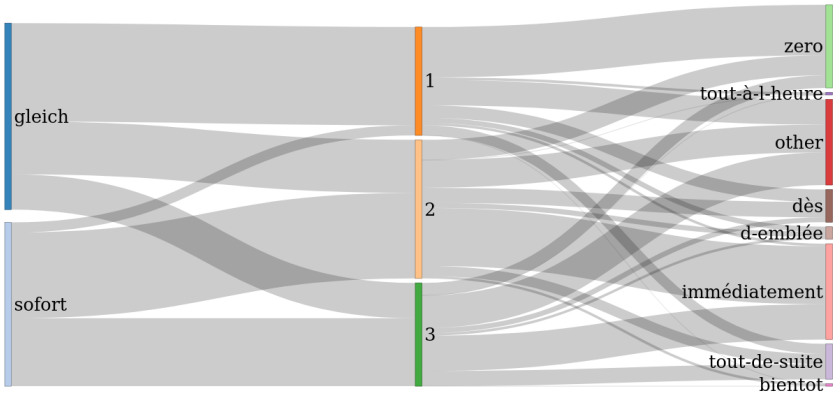



Figure 17: Comparative Sankey plot showing associations of IP-adverbials with French IP-adverbials, given a specific context cluster (solution with three clusters). The plot was generated with the package `networkD3` for `R` (Allaire et al. 2017).

quencies of combinations of levels for pairs of crossed variables. For example, it would be conceivable that most or even all occurrences of *sofort* going to 'deic-dir' actually end up at 'zero'.

Translation choices can be represented more precisely with visualizations that we call Context-Conditional Translation Graphs (CCTG; see also Gast & Atayan 2019). Consider the graph in Figure 18. It shows frequencies of choices made in German-to-French translation, under specific contextual conditions. The colours of the edges correspond to the three clusters of context variables (see also the caption of the graph). Figure 18 shows that *gleich* is indeed mostly not translated at all in Context 1. *Sofort* typically goes to *immédiatement* in Contexts 2 and 3. Unlike suggested by Figure 17, there does not seem to be any particular association between *gleich* and 'other', as the three arrows between these nodes have approximately the same width.

## 5.2 Bayesian networks

In Section 2 it was mentioned that choices in translation can be regarded as coming with a specific conditional probability. For example, we can compare the conditional probability of finding *immédiatement* in Context 2 if there is *gleich* in the German original with its conditional probability if there is *sofort* in the original. That would tell us something about the relationships between context types in the original, expressions in the original and expressions in the target language (note that translators may of course also choose to change the context

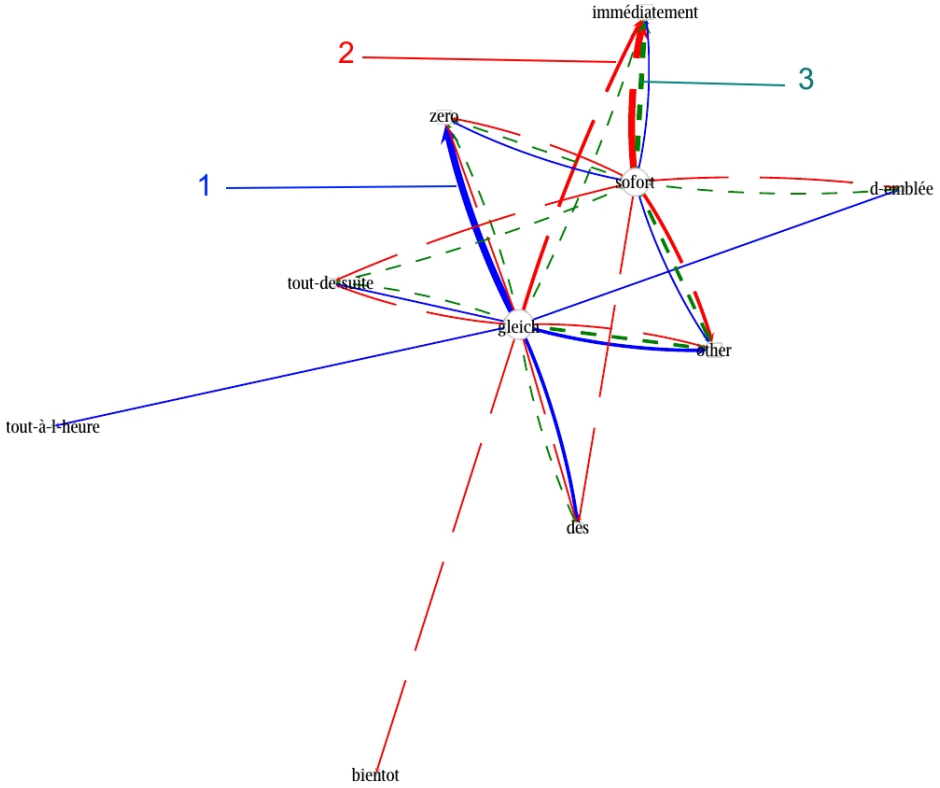



Figure 18: Context-Conditional Translation Graph showing French translation choices for German *gleich* and *sofort*, given specific context clusters. The (solid) blue edge indicates translation in Context 1, the (long-dashed) red line represents translations in Context 2, and the (short-dashed) green line corresponds to Context 3. The graph was generated with the package `graph_tool` for `Python` (https://graph-tool.skewed.de/static/doc/); see the script `IpNetwork.py` in the Supplementary Materials.

type, in which case we have to deal with yet another variable determining the probabilities of using a specific expression in the target language). While we cannot offer a complete and statistically sound analysis of this type, we will present a general framework which is flexible enough to accommodate such cases, and that makes use of visualizations, more specifically, graphical models.

Graphical models capture the type of complex data structure as encountered in translation studies in the form of networks or graphs representing dependencies between random variables (cf. Koller & Friedman 2009). We will use Bayesian networks, a type of graphical model that represents the (assumed, hypothesized)

causal relations between variables. We will not be concerned with the causal relationships between specific distributional variables. For example, it seems reasonable to assume that ILLOCUTIONARY FORCE has a causal effect on the variables AGENT and INTENTIONALITY; if a speaker wants an addressee to do something, that implies a certain amount of control on the part of the addressee. This way of causal modeling is not specific to the type of distribution under discussion in this section, i.e., choices made in a target language under the influence of a source language expression.

It should be noted that corpus linguistic methods are not adequate for investigating causal relationships in the process of bilingual language production. We would need experimental evidence for that. What we can do with corpus data, however, is model the cognitive processes during translation and test the predictions derived from this modeling. A simple example of a model of bilingual language production is shown in Figure 19 in the form of a directed acyclic graph (DAG).[9] It represents the choice of a French ImPost-adverbial in a translation from German. The nodes represent variables, and they are associated with local probability distributions:

The distribution of **Context types** is represented at the top of the graph. We assume the three types distinguished by the 'smaller' PAM-cluster solution (cf. Sect. 4). Each of the context types has a certain (relative) frequency. We estimated the frequencies of the three context types from our sample.

The specific context type conditions a frequency distribution in the originals. We can, again, determine this distribution on the basis of our sample. The node **OrigIPDe** is associated with conditional probabilities of the form $p(w_{de}|c)$ – the probability of using a specific German ImPost-adverbial$w_{de}$, given a context $c$.

The node **HypIpFr** stands for a hypothetically used French ImPost-adverbial – the ImPost-adverbial that *may* have been used in monolingual (French) language production, in the specific context. Like the node OrigIPDe, this node is associated with a conditional frequency distribution, $p(w_{fr}|c)$. This distribution can be estimated from the original French data.

Finally, the node **TrlIPFr** stands for the ImPost-adverbial used in the French translation from German. According to the DAG in Figure 19, this choice is conditioned by the probability of using a specific French expression (as in monolingual language production), and the influence from the German ImPost-adverbial, reflecting form-to-form associations in translation. The node TrlIPFr thus represents a distribution over French ImPost-adverbials, conditioned by the hypothetical French ImPost-adverbial and the German ImPost-adverbial in the original

[9] The graph was generated with the package `bnlearn` for `R`. In Python, the module `pgmpy` can be used for Bayesian networks.

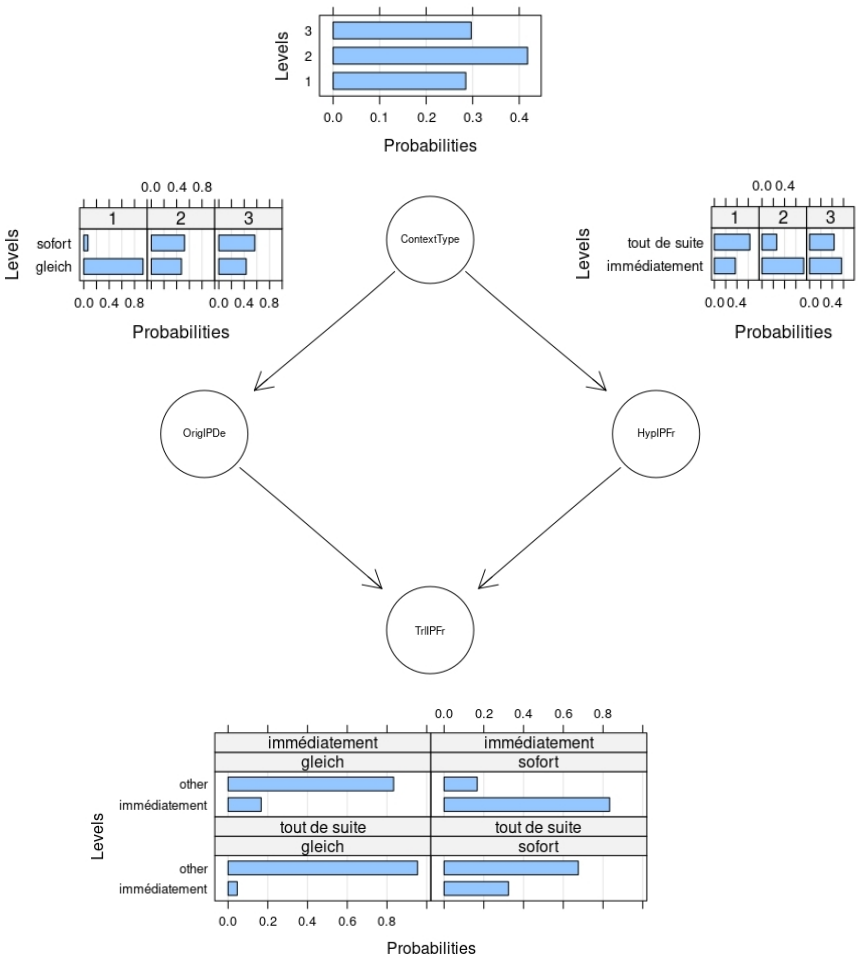



Figure 19: Directed acyclic graph for IP-adverbials (comparative/directional, with clustered observations). The graph was created with the package `bnlearn` for `R` (Scutari 2010).

Table 7: Empirical probabilities (left) and theoretical probabilities according to the Bayesian network (right) for the occurrence of *immédiatement* in French translations, in specific contexts, given a specific German ImPost-adverbial in the original.

| | empirical | | theoretical | |
|---|---|---|---|---|
| ContextType | *gleich* | *sofort* | *gleich* | *sofort* |
| 1 | 0 | 0.25 | 0.09 | 0.51 |
| 2 | 0.24 | 0.53 | 0.13 | 0.7 |
| 3 | 0.07 | 0.48 | 0.11 | 0.61 |

sentence. This distribution cannot be observed, as no (hypothetical) monolingual ImPost-adverbial is ever produced in French translation from German. In order to infer the hypothetical French expressions, we trained a classifier (a support vector machine/SVM) on all explanatory variables, predicting a given ImPost-adverbial for each translated French sentence.[10]

The network shown in Figure 19 represents a hypothesis about mediated language production. By breaking down the joint probability distribution into local, partially independent distributions, we can model components of the highly complex process of mediated language production. As our data set is quite small and the estimates for some parameters are not very precise, we will illustrate this procedure with a binary choice, by modeling the use of *immédiatement* in French translations. All French ImPost-adverbials other than *immédiatement* were subsumed under the category of 'other'.

The Bayesian network in Figure 19 allows us to determine theoretical distributions of French ImPost-adverbials under different conditions, and to compare them to the empirical distributions. This can give us an estimate of the plausibility of the causal model. Tables 7 shows the empirical probabilities on the left and the theoretical probabilities (according to the Bayesian network) on the right. While the ranks of the expressions – by context type and ImPost-adverbial – are identical, the two distributions are considerably different. Even though the model in the form displayed in Figure 19 may not be entirely realistic, it provides a starting point for a more elaborate model. This is one of the main assets of Bayesian networks: they can be extended and built up step by step, and their graphical nature helps researchers to develop an understanding of the data.

[10]The model has an accuracy of 88.2% for the entire training set, and of 79.5% with a 0.9/0.1 training/test split, averaging over ten-fold cross-validation. We used a radial kernel, with $C = 3$ and $\gamma = 0.1$.

## 6 Conclusions and outlook

The present chapter has provided an overview of various types of methods used in comparative corpus linguistics and translation and interpreting studies. Some of the methods are well known, – e.g., mosaic plots (as in Figure 1) – while others may be less familiar to parts of the community of contrastive linguists and translation scholars.

We discussed two major types of ordination: CA (with and without stacking), and MCA (with and without supplementary variables). We made an effort to convey an intuitive understanding of the mathematics underlying these techniques. Users have to make specific choices and decisions, e.g., CA vs. MCA, the type of map (symmetric, asymmetric, row principal, etc.), and such decisions need to be informed. We also highlighted the connection between (M)CA and the familiar Chi-squared statistic (e.g., the connections between this statistic and Chi-squared distance, inertia, and the $\phi$-coefficient).

It seems to us that graph-based methods have some potential that has not fully been exploited. Context-Conditional Correlation Graphs (CCCGs, cf. Sect. Fig. 4 in Sect. 3) allow us to represent a large amount of information about pairwise associations between variables in a single plot. Sankey diagrams (Figures 15, 16 and 17), which are commonly used in other fields, do not seem to be widespread in translation studies, though they offer a very good way of summarizing directional data. Translation graphs visualize choices made in translation, under specific conditions. While the type of graph that we provided (Figure 18 in Sect. 5) does not contain any type of statistical information, such information could be added, potentially using Bayesian statistical reasoning as outlined in Sect. 5.2, where we presented a type of graphical model, i.e., Bayesian networks.

Once again, we would like to remind readers that the purpose of the present chapter was not to present research results about ProxPost-adverbials. The data was mainly used for the purpose of illustration. As previously mentioned, more thorough studies on ProxPost-adverbials were presented in earlier publications of ours (Atayan et al. 2019, Gast & Atayan 2019, Gast et al. 2019, Atayan 2021a,b), where some of the visualizations presented in this chapter were also used.

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# Appendix A

## A.1 Notes on Sect. 3

When dealing with categorical data as discussed in this chapter, we typically want to know if two or more variables correlate. In other words, we want to know if a property of some observation – a level of a variable – allows us to make

inferences about some other property of that observation. According to the null hypothesis (of statistical independence), which says that the two variables are unrelated, the conditional probability of observing a specific property $A$ in some observation $x$, given some other property $B$ of $x$, is identical to the probability of observing $A$ in $x$ independently of $B$ (cf. (9a)). Put differently, the probability of any $x$ exhibiting both $A$ and $B$ is the product of the probabilities of $A$ and $B$ (cf. (9b)).

(9) Assumption of independence:
   a. $p(A|B) = p(A)$
   b. $p(A \wedge B) = p(A) \cdot p(B)$

If two variables do not correlate, the frequencies in a contigency table will be proportional along both dimensions. Consider Table 8.

Table 8: Schematic contingency table

| | A: level 1 | A: level 2 | Σ |
|---|---|---|---|
| B: level 1 | a | b | a+b |
| B: level 2 | c | d | c+d |
| Σ | a+c | b+d | a+b+c+d |

If the variables $A$ and $B$ are independent of each other, we expect the ratio of any two pairs of cells $\frac{c_1}{c_2}$ and $\frac{c_3}{c_4}$ to be the same if $c_1$ and $c_2$, and $c_3$ and $c_4$, are in the same column, and $c_1$ and $c_3$, and $c_2$ and $c_4$, are in the same row (including the marginal row and column sums). For instance:

(10)
   a. $\frac{a}{c} = \frac{b}{d}$
   b. $\frac{a}{a+c} = \frac{b}{b+d}$
   c. $\frac{a}{c} = \frac{a+b}{a+b+c+d}$
   d. $\frac{a}{b} = \frac{c}{d}$
   e. $\frac{a}{a+b} = \frac{c}{c+d}$
   f. $\frac{a}{a+c} = \frac{a+b}{a+b+c+d}$

Based on such analogies, we expect the following count for cell $a$, under the assumption of statistical independence of the variables. The equation in (11) follows from (10f) by multiplication with $(a+c)$ (the symbol $\hat{}$ on the $\hat{a}$ indicates that this variable represents an expected value).

(11) $\hat{a} = \frac{(a+c)(a+b)}{a+b+c+d}$

The difference between the observed frequency $a$ and the expected frequency $\hat{a}$ provides a way of determining deviations from statistical independence.

In what follows we use the matrix formalism, which provides a compact way of expressing and generalising the above example. $C$ represents the original contingency table, and $\hat{C}$ corresponds to the matrix of the expected frequencies as calculated from the independence condition as stated in 9. The matrix of **Pearson residuals** is then calculated as:

(12) Pearson residuals
$R = (C - \hat{C})/\sqrt{\hat{C}}$

The matrix $\sqrt{\hat{C}}$ contains the square roots of the cells in $\hat{C}$. Division by the square root of the expected value is a standardization step. We assume Poisson sampling (independent selection of subjects), so each cell is assumed to follow a Poisson distribution. In a Poisson distribution, the variance is equal to the expected value (both values are represented by the parameter $\lambda$). We divide the difference between the expected and the observed value by the square root of the variance – the standard deviation – to standardize the values.

In matrix notation, the matrix of expected values $\hat{C}$ can be represented as follows:

(13) a. $\hat{C} = (m_m \otimes m_n)/N$, (the outer product of row and column margins, divided by the number of observations)
b. $m_m = 1C, m_n = 1C^{\mathsf{T}}$, (row and column sums)
c. $N = 1^{\mathsf{T}}C1$ (the sum of all cells)

The **adjusted Pearson residuals** are obtained as shown in (14). We divide the difference between the observed and the expected value by an estimate of the standard error based on the assumption that the marginal sums are fixed, therefore assuming a hypergeomteric distribution. Adjusted Pearson residuals are always larger than unadjusted Pearson residuals, as the additional term in the demoninator – $\sqrt{(1 - m_m/N)(1 - m_n/N)}$ – is always smaller than 1.

(14) $\tilde{r} = \frac{C - \hat{C}}{\sqrt{\hat{C}(1 - m_m/N)(1 - m_n/N)}}$

## A.2 Notes on ordination

The information in this section is mainly intended to facilitate an intuitive understanding of the methods. For more complete and more accurate information, please consult Greenacre (2016).

### Some terms and concepts

A **contingency table** is a table with counts of observations with specific properties, cf. Table 9.

Table 9: Contingency table

| | comm | dir.boul | repr | Σ |
|---|---|---|---|---|
| *gleich* | 4 | 3 | 18 | 25 |
| *sofort* | 2 | 14 | 18 | 34 |
| *unverzüglich* | 1 | 6 | 2 | 9 |
| Σ | 7 | 23 | 38 | 68 |

A **percentage table** shows the relative frequency of each cell, i.e., the cell value divided by the number of observations (cf. Table 10). The **row** and **column masses** (the margins of Table 10) are the relative frequencies of all observations in specific rows or columns.

Table 10: Percentage table

| | comm | dir.boul | repr | Σ |
|---|---|---|---|---|
| *gleich* | 0.06 | 0.04 | 0.26 | **0.37** |
| *sofort* | 0.03 | 0.21 | 0.26 | **0.50** |
| *unverzüglich* | 0.01 | 0.09 | 0.03 | **0.13** |
| Σ | **0.11** | **0.34** | **0.56** | 1 |

The **profile** of a given category constituting the row of a contingency or percentage table is the rowwise proportion of that category (cf. Table 11).

Table 11: Row profiles

| | comm | dir.boul | repr | Σ |
|---|---|---|---|---|
| *gleich* | 0.16 | 0.12 | 0.72 | 1 |
| *sofort* | 0.06 | 0.41 | 0.53 | 1 |
| *unverzüglich* | 0.11 | 0.67 | 0.22 | 1 |

The **average profile** – geometrically speaking the **centroid** – is the average of the row profiles, weighted by the row masses. For the first column:

(15) $\hat{c_1} = 0.37 \cdot 0.16 + 0.5 \cdot 0.06 + 0.13 \cdot 0.11 = 0.1$

The average profile for Table 9 is thus:

(16) $\hat{C}_{j+} = (0.1\ 0.34\ 0.56)$

We can obtain a **standardized profile table** by scaling the columns of the profile table (Table 11) by the inverse of the square root of the average profile values (i.e., we divide each value by the square root of the corresponding average profile value). For example, the average profile value of the first column is 0.1, and its square root is ≈ 0.32. If we divide 0.16 (the first value of the profile of *gleich*) by 0.32, we obtain a value of 0.5:

Table 12: Standardized profile table

| | 1 | 2 | 3 |
|---|---|---|---|
| gleich | 0.5 | 0.21 | 0.96 |
| sofort | 0.18 | 0.71 | 0.71 |
| unverzüglich | 0.35 | 1.15 | 0.30 |

In a coordinate system scaled by the inverse of the square root of the expected value Euclidean distances are equivalent with Chi-squared distances.

The **Chi-squared distance** $\chi^2_\Delta$ is the square root of the sum of the squared differences between the observed values and the expected values, divided by the expected values. For each row $m$ of a matrix $C_{m\times n}$:

(17) $\chi^2_{\Delta,m} = \sqrt{\sum_{i=1}^{n} \frac{(C_{m,i} - \hat{C}_{m,i})^2}{\hat{C}_{m,i}}}$

($\hat{C}$ being the matrix of expected values, see (13))

The sum of the Chi-squared distances of a matrix provides the basis for the concept of **inertia** in Correspondence Analysis. The inertia of a contingency table is "the weighted average of the squared $\chi^2$ -distances between the row profiles and their average profile" (Greenacre 2016: 29). It is weighted by the row masses $M$. For a matrix $C_{m\times n}$,

(18) $I_C = \sum_{i=1}^{m} \sum_{j=1}^{n} M_i \frac{(C_{i,j} - \hat{C}_{i,j})^2}{\hat{C}_{i,j}}$.

Note that the Chi-squared distance is identical to the Chi-squared value of a table, divided by the number of observations, and that it is equivalent to the $\phi$-coefficient squared, so $\phi = \sqrt{\chi^2/n}$.

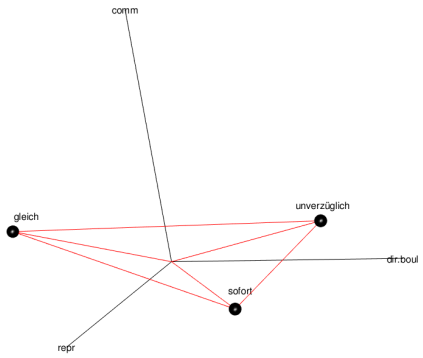



Figure 20: Visualizing the standardized residuals of Table 1

**Singular value decomposition**

As a starting point of singular value decomposition, we use a table of standardized residuals. It is obtained just as was in described in Sect. A.1 for the Pearson residuals. However, in Correspondence Analysis we calculate standardized residuals from a percentage table as shown in Table 10, not from a contingency table as shown in Table 9. The two tables differ by a factor of $\sqrt{N}$, i.e., in our case, 8.246 ($=\sqrt{68}$). Table 13 contains standardized residuals.

Table 13: Table of standardized residuals

| | comm | dir.boul | repr |
|---|---|---|---|
| *gleich* | 0.11 | -0.23 | 0.13 |
| *sofort* | -0.10 | 0.09 | -0.03 |
| *unverzüglich* | 0.01 | 0.21 | -0.16 |

We can represent Table 13 in a three-dimensional coordinate system as shown in Figure 6 in the main text, which is repeated for convenience in Figure 20. Alternatively, we can think of Table 13 as a mapping from one three-dimensional coordinate system to another three-dimensional coordinate system. Let us assume a source system that consists of three unit vectors only, i.e., a standard basis (cf. Figure 20). When regarded as a transformation map, Table 13 (via matrix multiplication) maps the standard basis to the residuals shown in red in Figure 21. What matters here is that by regarding the matrix as a transformation map we can not only map the three standard vectors to new coordinates, we can map *any* point in the source system to the target system. For example, the point located at (0.5 0 0), which is halfway between the origin and the unit vector (1 0 0) in

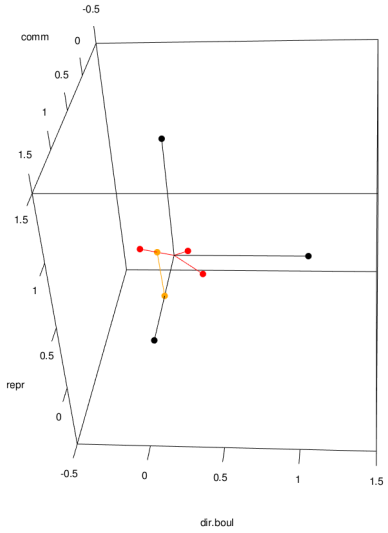



Figure 21: Table 13 as a mapping from a standard basis to a new coordinate system

the original system, will be mapped to a point that is halfway between the origin and the target vector of (1 0 0) in the new system (see the two orange points in Figure 21).

Under the transformation view of the matrix in Table 13 we can perform a 'global' transformation of the original coordinate system. Consider Figure 22, which shows a random sample of vectors (represented as points) of unit length. The cloud of points shown in Figure 22 can be mapped to the space of standardized residuals in the same way in which we mapped the three unit vectors to that space in Figure 20 with matrix multiplication. All points will be mapped to a 'disc' that contains the origin, see Figure 23.

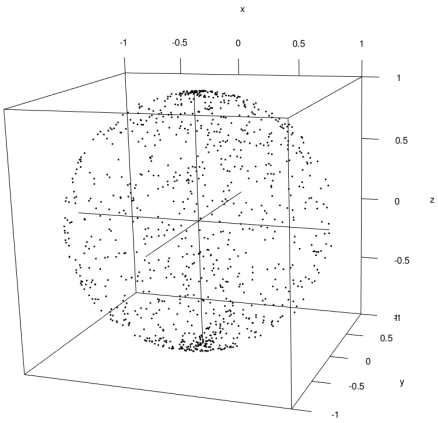



Figure 22: A random sample of 1,000 vectors of length 1

We can now look at the 'disc' of Figure 23 from a different point of view, see Figure 24. This plot shows two straight lines. The horizontal line points in the direction in which the points of the new cloud are spread out most. It is the first **singular vector**. The vertical line is orthogonal to the horizontal line. It shows the direction in which the data is spread out most, on the condition of being orthogonal to the first line. (Note that there appears to be only one line that is orthogonal to the first line, but remember that the disc is actually located in three-dimensional space [see Figure 23], so there are in fact an infinite number

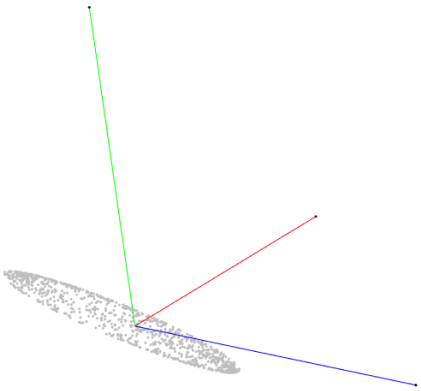


Figure 23: The random sample of 1,000 vectors from Figure 22 mapped to residual space

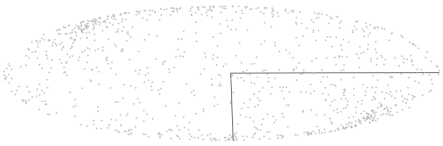


Figure 24: The random sample of 1,000 vectors from Figure 22 mapped to residual space, from a different perspective and with singular vectors

of vectors that are orthogonal to the first singular vector, pointing away from the disc).

Singular value decomposition is a method to find singular vectors and their lengths, the singular values. It decomposes a matrix $M$ into two square matrices $U$ and $V$ and a (rectangular diagonal) scaling matrix $\Sigma$ such that:

(19) $M = U\Sigma V^*$

where $^*$ denotes the conjugate transpose of a matrix. For matrices of the type dealt with in the present context, the conjugate transpose is identical to the transpose, i.e., $V^* = V^\mathsf{T}$.

The matrices $U$ and $V$ contain orthonormal bases as row and column vectors, which means that these vectors are orthogonal to each other, and each vector is of unit length. In the case of $U$, the row vectors are orthogonal, in the case of $V$, it is the column vectors that are orthogonal, hence:

(20) $UU^T = I$

(21) $V^TV = I$

In our case, the SVD delivers the following factorization (obtained with the function `svd` in R):

(22)
$$U \quad \begin{bmatrix} -0.72 & 0.35 & 0.61 \\ 0.28 & -0.65 & 0.71 \\ 0.64 & 0.68 & 0.36 \end{bmatrix} \qquad \Sigma \quad \begin{bmatrix} 0.39 & 0.00 & 0.00 \\ 0.00 & 0.12 & 0.00 \\ 0.00 & 0.00 & 0.00 \end{bmatrix} \qquad V \quad \begin{bmatrix} -0.25 & 0.91 & -0.32 \\ 0.81 & 0.02 & -0.58 \\ -0.52 & -0.41 & -0.75 \end{bmatrix}$$

To see that the first two row vectors of U are orthogonal (they stand in a $90^o$ angle to each other), consider their dot product:

(23)
$$\begin{aligned} & u_1 \cdot u_2 \\ & = (-0.715 \cdot 0.285) + (0.348 \cdot -0.647) + (0.606 \cdot 0.707) \\ & = -0.204 - 0.225 + 0.429 = 0 \end{aligned}$$

To see that the length of the first vector is 1:

(24) $\sqrt{-0.715^2 + 0.348^2 + 0.606^2}$
$= \sqrt{0.511 + 0.121 + 0.367}$
$= \sqrt{1} = 1$

The rows of $U$, and the columns of $V$, constitute the bases for the coordinates of the rows and columns in CA. Both matrices are scaled by the inverse of the square root of the average profile values ($W_m, W_n$ below). The matrix of interest is moreover scaled by the singular values in matrix $\Sigma$. If we want to locate rows in a coordinated system defined by their associations with columns, we will only scale the left singular values, i.e., matrix $U$, by $\Sigma$. If we want to locate columns in a coordinate system defined by their associations with rows, we scale the right singular values, i.e., matrix $V$, by $\Sigma$. Sets of coordinates, here represented as $C$, can thus be obtained as follows (in the text we were only interested in the row analysis):

(25) row analysis
a. $C_{row} = W_m U \Sigma$
b. $C_{col} = W_n V$

(26) column analysis
a. $C_{col} = W_n V \Sigma$
b. $C_{row} = W_m U$

In an asymmetric map, where rows are represented in principal coordinates and columns in standard coordinates, we can reconstruct the original matrix from the map via the inner product between row and column vectors. For example, the ImPost-adverbial *gleich* and the column 'repr' have the following coordinates:

(27) *gleich*: (−0.46 0.07)

(28) 'repr': (−0.70 −0.54)

These coordinates first need to be scaled by the square root of the row masses/ average profile (to 'undo' the scaling that was applied previously):

(29) *gleich*: 0.606 (−0.463 0.067) = (−0.281 0.041)

(30) 'repr': 0.748 (–0.702 −0.545) = (–0.525 −0.407)

The inner product of these vectors is:

(31) $-0.281 \cdot -0.525 + 0.041 \cdot -0.407 = 0.131$

The result, 0.131, is the standardized residual of the cell for *gleich* and 'repr'.

## A.3 Notes on Sect. 4

Figure 25 shows the silhouette plot for the clustering reported on in Sect. 4.

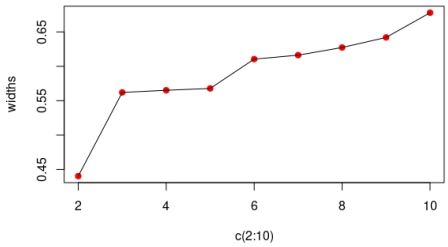



Figure 25: Silhouette widths for cluster solutions